

Outlines

Outlines → make writing more efficient

- Partially answers “what do I want to say?”
- Clarifies “work still to be done” and gives you a long lead time to get it done
- Can get high-value input from collaborators at an early stage.
- Reveals structural problems *early*, before investing time writing.

What do I mean by “structural problems”?

Are you....

Explaining things in the wrong order?

Explaining too much at the same time?

Missing information?

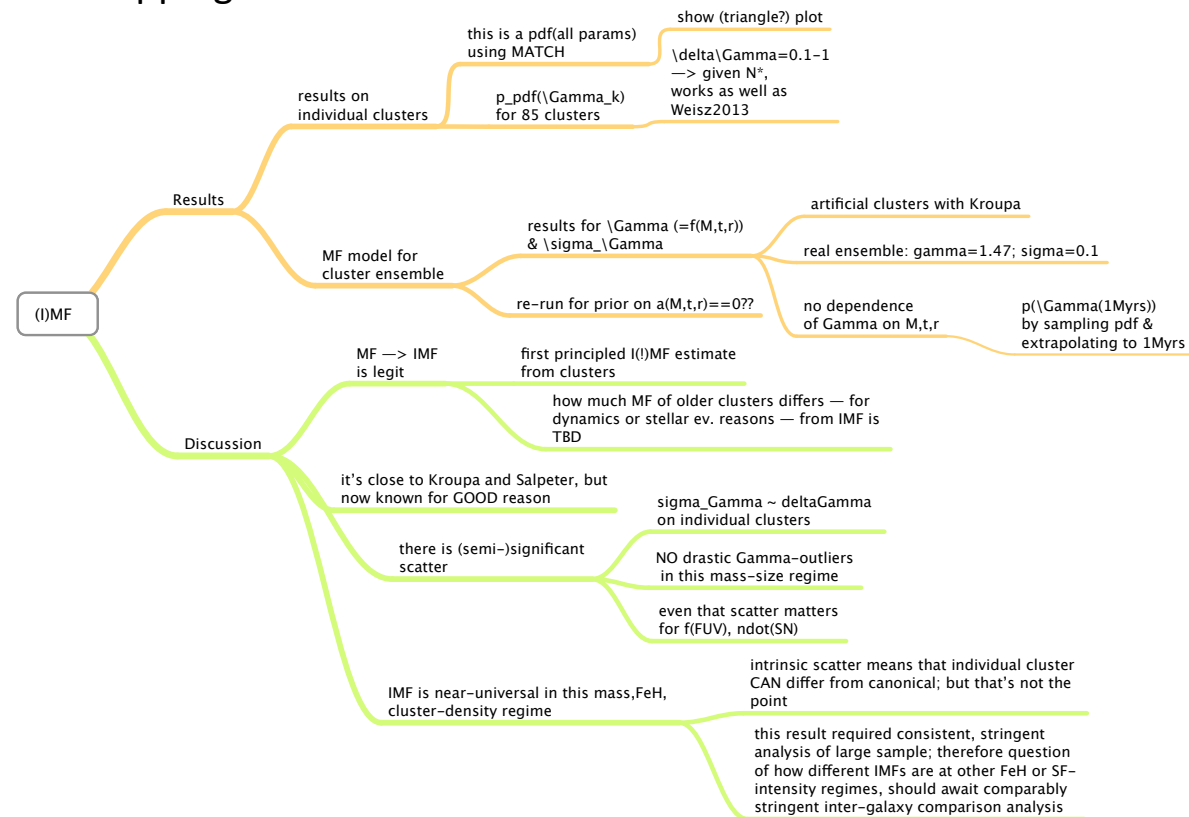
Motivation unconvincing?

Line of argument confusing?

Structural Problems

- You can't just tweak words and sentences to fix these.
- You have to move or cur entire paragraphs and sections.
- Easy to fix EARLY before text is written.

Mind Mapping



<http://mindnode.com>

Has free trial. Paid version lets you drag & drop nodes.

Outline for Science Justification

Abstract: Summarizes the entire proposal

Section 1: Facts and summary of Sections 2-4

Section 2: The Problem/Motivation/Story

Section 3: The Goal & Strategy. How you will use “key component” + “Target” to solve the Problem

Section 4: Importance of the solution to the subfield & Broader Impacts (legacy value)

Section 2: The Problem

Abstract Outline

Facts

Importance of Facts

Impact within Sub Field

Problem

Goal – identify the “key component” that will solve the problem

We propose to...

Strategy – to utilize/generate the key component ,
(Justify HST/JWST, explain utilizing the “target”)

Suitability

Importance of Solution

Impact within Sub Field

Broader Impact

Out-of-field Impact

New Models of the Milky Way's Dark Matter Distribution for the Era of High Precision Astrometry

Understanding the assembly history and dark matter distribution of our Milky Way (MW) is a major challenge for astrophysics. Thanks to the unique capabilities of HST, proper motions of satellite galaxies, globular clusters and stellar streams have been measured with accuracies of order ~ 0.05 mas/yr (~ 10 km/s) at distances of 50-300 kpc. When combined with detailed models of the MW's halo potential, such measurements become high-precision tools to constrain the dark matter mass profile of the MW and compute accurate orbital histories of satellites. However, the MW hosts a pair of massive dwarf galaxies, the LMC and SMC, that contribute to its dark matter distribution and change the shape of the potential in a non-symmetrical, time evolving manner. To date, these effects have not been accounted for in existing models of the MW halo. We propose to develop high resolution simulations to quantify the time evolving structure of the MW's dark matter halo owing to the influence of the LMC and SMC. These novel models will enable rapid orbital integration of halo objects (satellites, globular clusters, stellar streams), using high accuracy HST proper motions, while also capturing the complex halo potential resulting from the LMC-SMC-MW interaction. The era of high-precision astrometry has arrived, yet we do not currently have an appropriate theoretical framework to study the assembly history of MW-like galaxies in the presence of massive satellite perturbers. Our proposed program is thus critical to ongoing HST programs and all efforts to understand the structure and evolution of the dark matter halo of our Galaxy and analogous systems like M31 and its massive satellite, M33.

Facts

Importance of Facts

Problem

Proposal

Strategy - Soln

Importance of Solution

Broader Impact

- Review the Section 2 example
- Answer the outline prompts (copy and paste the relevant portions of the text)
- Discuss with neighbor(s)

Section 3: The Strategy

Nathan Smith
Very small proposal

Spectroscopy of the complex core of VY Canis Majoris

VY Canis Majoris (VY CMa) is the most luminous, largest, and presumably the initially most massive red supergiant (RSG) known. It is surrounded by a spectacular asymmetric nebula that was imaged by HST over 20 years ago, and shows a complex array of knots, arcs, and large-scale axisymmetric structure. It is a key object

for understanding the extreme mass loss of the most massive RSGs and a benchmark for progenitors of SNe that interact with dense circumstellar material.

Spectra reveal a range of outflow velocities in these structures that are faster than typical RSG winds, suggesting blasts of uneven mass-loss episodes over the past 1000 yr. The origin of this instability is not known, but it is usually attributed to either magnetic/convective instability near the Eddington limit, or to binary interaction with an unseen companion star. Although the central source in the nebula appeared unresolved in the earlier HST/WFPC2 imaging, a recent ground-based extreme-AO image has revealed that the central core of the nebula is not a point source. Instead, the core seems to be bisected by a dark obscuring lane (a disk?), and a fainter point source (a companion star?) is resolved at about 120 mas from the core. These features would have been just at the edge of the resolution limit in dithered WFPC2 images.

We propose to obtain STIS spectra of the core of VY CMa in order to (1) record the spectrum of the fainter companion candidate, to see if it is reflected RSG light or a companion star, and (2) obtain the spectrum across the central obscuring lane to determine if it is molecular/dust obscuration and estimate its column density.

Facts

Importance of Facts

Problem

Proposal

Strategy - Soln

Importance of Solution (?)

Broader Impact (??)

- Review the Section 3 example
- Answer the outline prompts (copy and paste the relevant portions of the text)
- Discuss with neighbor(s)